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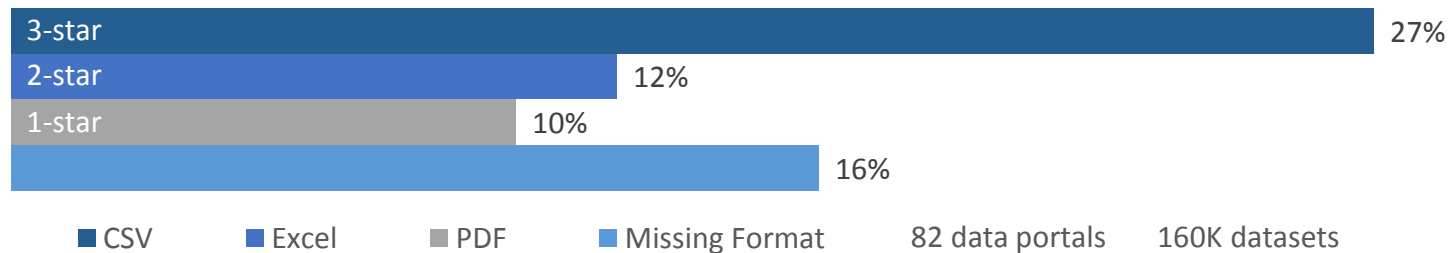
Improving Open Data Usability through Semantics

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Problem

- Open Data is (partially) structured and tabular [1]:



- Use of existing Semantic Web technologies to improve existing 3-star Open Data
 - Integrate and interlink data
 - Allows cross-portal integration
- Focus on tabular data

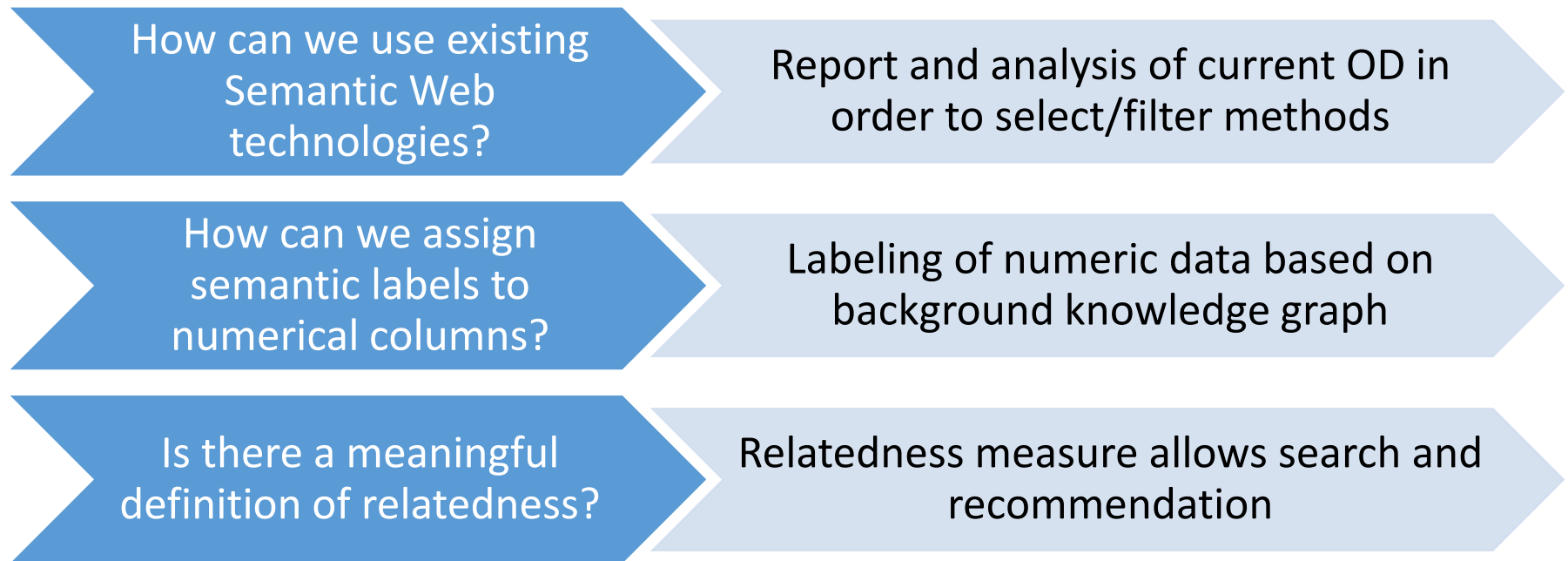
State of the art

- Existing work mainly use named-entity recognition techniques
- Web/HTML tables differ from typical Open Data tables:
 - Domain:** e.g., public administration data, statistical data, weather data, elections, ...
 - Structure:** OD tables contain large amount of numerical columns

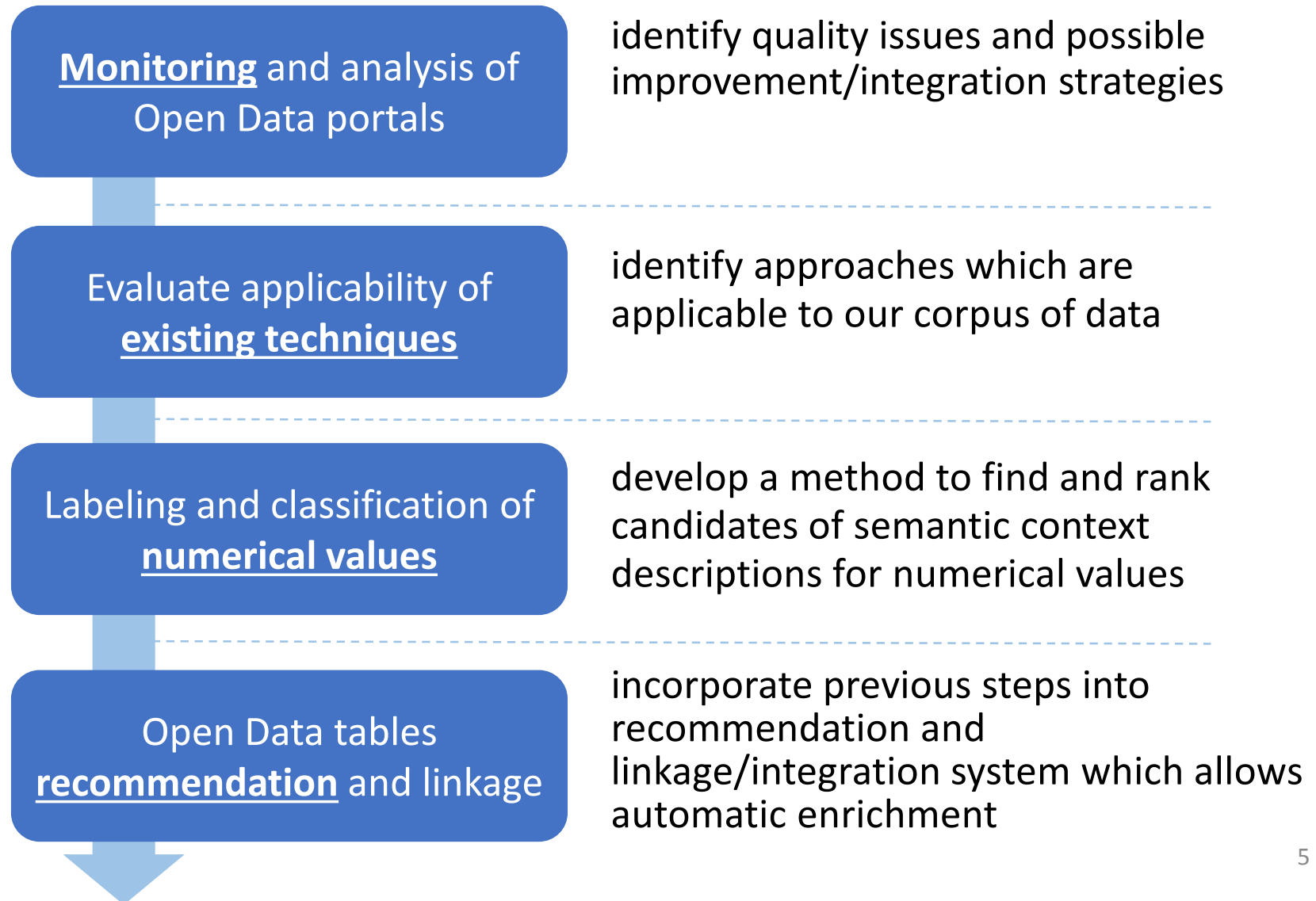
Rank	Previous Rank	Name	Country	Ranking Points	Wohnungen in den 250 Zaehlbezirken in Wien - Registerzaehlung 2011 Housing units in 250 st					
1	1	Novak Djokovic	Serbia	16,950	NUTS1	NUTS2	NUTS3	DISTRICT_CODE	SUB_DISTRICT_CODE	WHG_TOTAL
2	2	Andy Murray	Great Britain	8,915	AT1	AT13	AT130	90100	90101	3004
3	3	Roger Federer	Switzerland	6,425	AT1	AT13	AT130	90100	90102	1049
4	4	Rafael Nadal	Spain	5,335	AT1	AT13	AT130	90100	90103	1389
5	5	Stanislas Wawrinka	Switzerland	5,035	AT1	AT13	AT130	90100	90104	1014
6	6	Kei Nishikori	Japan	4,155	AT1	AT13	AT130	90100	90105	1337
7	7	Milos Raonic	Canada	3,175	AT1	AT13	AT130	90100	90106	1915
8	8	Dominic Thiem	Austria	3,175	AT1	AT13	AT130	90100	90107	2032
					AT1	AT13	AT130	90200	90201	5178
					AT1	AT13	AT130	90200	90202	6345
					AT1	AT13	AT130	90200	90203	7549
					AT1	AT13	AT130	90200	90204	8388
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					AT1	AT13	AT130	90200	90206	4237
					AT1	AT13	AT130	90200	90207	7812
					AT1	AT13	AT130	90200	90208	1478
					AT1	AT13	AT130	90200	90209	7547

Research Question & Hypothesis

The semantic analysis of Open Data CSVs increases the usability of Open Data and ideally generates 5-star linked data.



Approach



Preliminary results

- **Monitoring and analysis of Open Data portals [2,3]**
 - Periodical monitoring and assessment
 - (Meta)data quality metrics, e.g.: Retrievability, existence of contact or license information
 - <http://data.wu.ac.at/portalwatch>
- **Labelling of numerical values [4]**
 - Assign multi-level semantic labels to numerical values
 - Based on knowledge graph constructed from DBpedia

[2] Neumaier, S., Umbrich, J., Polleres, A.: Automated quality assessment of metadata across open data portals. ACM Journal of Data and Information Quality (JDIQ), 2016.

[3] Mitlöhner, J., Neumaier, S., Umbrich, J., Polleres, A.: Characteristics of Open Data CSV Files. In: 2nd International Conference on Open and Big Data (August 2016)

[4] Neumaier, S., Umbrich, J., Parreira, J.X., Polleres, A.: Multi-level semantic labelling of numerical values. In: The 15th International Semantic Web Conference. Kobe, Japan (October 2016)

Evaluation plan

- Existing Semantic label annotation and linking evaluation is based on (manually created) Gold standard datasets:
 - Crawl of the Web [5], Domain specific datasets (e.g., IMDB, MusicBrainz)
- Our numeric labelling approach:
 - Cross-validating over a sample of DBpedia
 - “In the wild” on tabular data extracted from Open Data portals to gain insights and report challenges
- Test-data corpus for further evaluations:
 - CSV data monitored by Open Data Portal Watch

Reflections

- **Automated integration** and high-quality linkage of all Open Data CSV tables **not possible**
- **Partial mappings** already generate high value and increases the usability of current Open Data
- ADEQUATe project [6]:
 - “Analytics & Data Enrichement to improve the Quality of Open Data”
 - Identified the potential (and demand) for integration and semantic labelling of Open Data resources



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